



Genome-wide identification, systematic analysis and characterization of *SRO* family genes in maize (*Zea mays* L.)

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Abstract

SIMILAR TO RCD ONE (SRO) is a small plant-specific gene family, which play essential roles in plant growth and development as well as in abiotic stresses. However, the function of SROs in maize is still unknown. In our study, six putative *SRO* genes were isolated from the maize genome. A systematic analysis was performed to characterize the *ZmSRO* gene family. The *ZmSRO* gene family was divided into two groups according to the motif and intron/exon analysis. Phylogenetic analysis of them with other plants showed that the clades of SROs along with the divergence of monocot and dicot and *ZmSRO*s were more closely with *OsSRO*s. Many abiotic stress response and hormone-induced *cis*-regulatory elements were identified from the promoter region of *ZmSRO*s. Furthermore, RNA-seq analysis indicated that *SRO* genes were widely expressed in different tissues and development stages in maize, and the expression divergence was also obviously observed. Analyses of expression in response to PEG6000 and NaCl treatment, in addition to exogenous application of ABA and GA hormones showed that the majority of the members display stress-induced expression patterns. Taken together, our results provide valuable reference for further functional analysis of the *SRO* gene family in maize, especially in abiotic stress responses.

Keywords Systematic analysis · Maize · *SRO* · Abiotic stress

Introduction

Plant-specific genes are essential for maintaining plant characteristics, adapting to environmental changes and supporting the growth and development of the plants. Up to now, a lot of plant-specific genes have been identified from different plants; however, the functions of many of them are still unknown (You et al. 2014, 2015; Gutierrez et al. 2004; Ning et al. 2017). Recently, as a small group of plant-specific protein family, the SIMILAR TO RCD ONE (SRO) protein family was identified from the plants (Jaspers et al. 2010). The SRO proteins contain a poly (ADP-ribose) polymerase catalytic (PARP) domain and a RCD1-SRO-TAF4

(RST) domain in the C-terminal region (Jaspers et al. 2009). Additionally, a WWE domain is also found in the N-terminal sequence of some SRO proteins (Jaspers et al. 2009, 2010; Aravind 2001). PARPs regulate many biological processes of plants, including programmed cell death (PCD), DNA damage and DNA repair, gene transcription, cell cycle and stress tolerance (Briggs and Bent 2011; De Block et al. 2005; Lamb et al. 2012; Vanderauwera et al. 2007). The WWE domain and RST domain are involved in protein–protein interactions (Jaspers et al. 2010; Liu et al. 2014). RADICAL-INDUCED CELL DEATH1 (RCD1) was the first identified plant SRO protein from *Arabidopsis*, and the structure prediction revealed that it contains a WWE domain and a RST domain (Ahlfors et al. 2004; Sundaram et al. 2008). The RCD1 not only participates in the stress response to salt, osmotic and oxidative, but also is involved in plant development by mediating the hormonal signaling such as SA and ABA (Overmyer et al. 2000; Teotia and Lamb 2009, 2011; Jaspers et al. 2010; Zhu et al. 2013).

Many of SROs have been isolated and characterized from different plants, such as *Arabidopsis* (Fujibe et al. 2004), rice (You et al. 2013), wheat (Liu et al. 2014), tomato (Babajani et al. 2009), cotton (Lv et al. 2017), *Brassica napus* (Anjum

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et al. 2015) and apple (Li et al. 2017b). The increasing researches indicate that *SRO* genes in plants are critical for plants' growth and development particularly under abiotic stress conditions. In *Arabidopsis*, the *SRO* protein family has six members, including *RCD1* and *SRO1* to 5. Both *AtRCD1* and *AtSRO1* with WWE domains are homologues, and own redundant roles in plant growth and development (Fujibe et al. 2004, 2006; Teotia and Lamb 2009; Teotia et al. 2010; Li et al. 2013). Double mutant of *AtRCD1* and *AtSRO1* displayed severe defects in embryogenesis and postembryonic development (Teotia and Lamb 2009, 2011). In addition, the functional mutant *sro1-1* showed resistance to the apoplectic ROS and salt stress, while *rcd1-3* showed the sensitivity phenotype, but both of them were resistant to osmotic stress in *Arabidopsis* (Teotia and Lamb 2009; You et al. 2013). Overexpression of *AtSRO5* and tomato *Sl-SRO11* in *Arabidopsis* could increase the salt tolerance of transgenic plants through reducing the level of H_2O_2 in roots (Babajani et al. 2009). In rice, five *SRO* genes (*OsSRO1a* to *OsSRO1e*) have been identified. The response to drought, salt, cold, heat, UV, wound, submerge and ABA treatment were characterized, which showed that all of them could be induced by drought, salt, heat and ABA treatment (You et al. 2014). In another study, *OsSRO1c*, as a direct target of the drought stress-related transcription factor *SNAC1*, could increase the tolerance of rice in drought and oxidative stress (You et al. 2013). In yeast, *OsSRO1a* interacting with *OsRBD1* protein, enhances the tolerance to salinity, osmotic and methylglyoxal stresses (Sharma et al. 2016). In addition, overexpression of the wheat *SRO1* gene in wheat and *Arabidopsis* can promote plants' growth and increase abiotic stress tolerance (Liu et al. 2014).

Maize (*Zea mays* L.) is an important cereal crop, which has been widely used in the study of genetics, evolution and other basic biological processes (Zhao et al. 2011). Although *SRO* genes have been widely investigated in *Arabidopsis*, rice, wheat and other species (Ahlfors et al. 2004; Jaspers et al. 2009; You et al. 2013; Anjum et al. 2015), the characteristics of each family member in maize remain poorly known. The published genomic information provides an excellent opportunity for genome-wide analysis of the *ZmSRO* gene family of maize (Schnable et al. 2009; Sekhon et al. 2011; Wang et al. 2015). Here, six non-redundant *ZmSRO* genes were identified and characterized in the maize genome. Their expression patterns were detected in response to various abiotic stresses, including salt, drought, ABA and GA. Our results provide a better functional understanding of the *ZmSRO* gene family of maize in the future.

Materials and methods

Identification of *ZmSRO* genes

To screen the *ZmSRO* genes in maize, the genome sequences of maize inbred line B73 were downloaded from <http://www.maizesequence.org/index.html> (Wang et al. 2015). The Hidden Markov Model (HMM) profile of the PARP domain (PF00644) and RST domain (PF12174) were obtained from the Pfam database (<http://pfam.sanger.ac.uk/>) (Finn et al. 2006). The HMM profile was used as a query to identify all sequences containing PARP or RST domain in maize against the maize protein database (p value = 0.001). The candidate protein sequences were further identified using Pfam and SMART (<http://smart.embl-heidelberg.de/>) (Letunic et al. 2009). All of the candidate sequences were aligned using ClustalW and checked manually to remove potentially redundant genes. The non-redundant genes named *ZmSRO1* to *ZmSRO6* were used for further analysis. Information about the number of amino acids and chromosome locations were obtained from the inbred line B73 sequence database. The molecular weight (kDa), isoelectric point (PI) and the grand average of hydrophobicity (GRAVY) of each deduced *ZmSRO* protein were calculated using the ExPASy program (<http://www.expasy.org/tools/>).

Gene structure and conserved motif analysis

The exon/intron structure of *ZmSROs* were determined by comparison of cDNAs with their genomic sequences using the Gene Structure Display Server (GSDS) program (<http://gsds.cbi.pku.edu.cn/index.php>) (Hu et al. 2015). The conserved motifs of the full-length amino acid sequences were detected using the Multiple Em for Motif Elicitation (MEME) web tool, (<http://meme-suite.org/tools/meme>) with the following parameters: optimum motif width ≥ 10 and ≤ 150 , and the maximum motif number set to 20.

Subcellular localization prediction and *cis*-regulatory elements analysis

Two online tools WoLF PSORT (<https://wolfpsort.hgc.jp/>) and Plant-mPLoc (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/>) were utilized to predict the subcellular localization of *ZmSRO* proteins (Chou and Shen 2010). The putative *cis*-regulatory elements of *ZmSROs* were identified from the 2000 bp upstream sequences of the start codon using the PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) (Lescot et al. 2002; Zhao et al. 2011).

Sequence alignment and phylogenetic analysis

The SRO protein sequences from *Arabidopsis*, rice and other plants were collected from the NCBI, RiceData and Phytozone. First, the complete amino acid sequences were aligned using ClustalX (1.83). Then with the Maximum Likelihood method, an un-rooted phylogenetic tree was constructed with 1000 bootstrap replicates in the software MEGA 6.0. The Dayhoff model was used to calculate the evolutionary distances (Hasegawa and Fujiwara 1993). AtPARP2 (accession number: AT4G02390) was used as an outgroup.

Transcriptional analysis of *ZmSROs*

The RNA-seq data were downloaded from the published paper by Stelpflug et al. (2016). Based on the sequenced ID of each *ZmSROs*, the transcriptional data from different tissues and development stages of 79 samples were obtained and used for RNA sequencing analysis. The expression data were normalized according to the row to generate a heatmap by the OmicShare platform (<http://www.omicshare.com/tools>).

Plant materials, growth conditions and stress treatment

The maize inbred line B73 was used in this study and the maize plants were grown in the growth chamber at 26 °C with a 16 h light/8 h dark cycle. The four-leaf stage (V4) seedlings were treated with abiotic stresses. The processing concentration and sampling time of each treatment were referred to the previous studies (You et al. 2014; Wang et al. 2015). For drought and salt stress, the seedlings were irrigated with 20% PEG 6000 and 200 mM NaCl solutions, respectively. For exogenous phytohormone treatments, the freshly prepared working solution of 100 µM abscisic acid (ABA) and 100 µM gibberellin A3 (GA3) were sprayed on the leaves, respectively. The samples were collected at 0, 0.5, 2, 4, 6, 8 h under drought and salt stress treatment and the samples were treated with ABA and GA3 were obtained at 0, 1, 3, 6, 9, 12 h. The time point of 0 h was set as control. Each sample was collected from five different maize seedlings as biological replicates. All the harvested samples were rapidly frozen in liquid nitrogen for RNA extraction and gene expression analysis and each assay was repeated three times in technology.

RNA extraction and cDNA synthesis

Total RNAs were extracted from the prepared samples of maize using TRIzol® Reagent (Invitrogen) according to the standard manufacturer's instructions, and then were digested using RNase-free DNase (TaKaRa) to remove genomic

DNA before reverse transcription. The quality of the isolated RNA was verified by 1% agarose gel electrophoresis and the concentration was quantified by spectrophotometric measurement (ND-1000 NanoDrop Technologies). The first-strand cDNA was synthesized from 2 µg of total RNA using a PrimeScript RT reagent Kit (TaKaRa) according to the manufacturer's instruction.

Quantitative real-time PCR

Quantitative real-time PCR (qPCR) assays were completed on the ABI 7300 Real-time PCR System. The qPCR reaction was conducted in 25.0 µl volumes, which contained 12.5 µl of 2×SYBR Premix ExTaq II (TaKaRa), 1 µl of each specific PCR primer, 1 µl of cDNA as the template, and 9.5 µl sterilization of ddH₂O. The amplification procedure was carried out as follows: 95 °C for 3 min, and 40 cycles (95 °C for 10 s, 57 °C for 20 s, and 72 °C for 20 s). The maize *Actin1* gene (accession number: GRMZM2G126010) was used as an internal control. Each sample was performed with three replicates. The relative of expression levels were calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen 2001). The primers used for qPCR in this study were listed in Table S1.

Results

Identification of the SRO family in Maize

The Hidden Markov Model (HMM) profiles of the PARP domain and RST domain were used to identify the putative SRO genes in the maize genome by the BLASTP program. Six non-redundant sequences were identified and confirmed by Pfam and SMART. The six SRO genes were assigned names *ZmSRO1* to 6 according to their chromosomal positions and number of amino acids, respectively. Among the six sequences, three of them (*ZmSRO1*, *ZmSRO3* and *ZmSRO6*) contain a WWE domain (Fig. 1a). Of the six genes, there were two on chromosome 1 and chromosome 5, and the remaining two were on chromosome 4 and 9, respectively. The length of six *ZmSRO* proteins ranged from 404 (*ZmSRO3*) to 596 (*ZmSRO5*) amino acids (aa) and the PI of them varied from 7.01 to 9.17 (Table 1). The GRAVY of the *ZmSRO* proteins varied from −0.364 to −0.465, showing that they were all hydrophilic proteins (Table 1).

Gene structure and conserved motif analysis of *ZmSROs*

The exon/intron analysis was performed against the maize genomic sequences of *ZmSROs*. *ZmSRO2*, 4 and 5 have a similar exon/intron structure with 6 exons and 5 introns, *ZmSRO1* and *ZmSRO6* have another similar structure

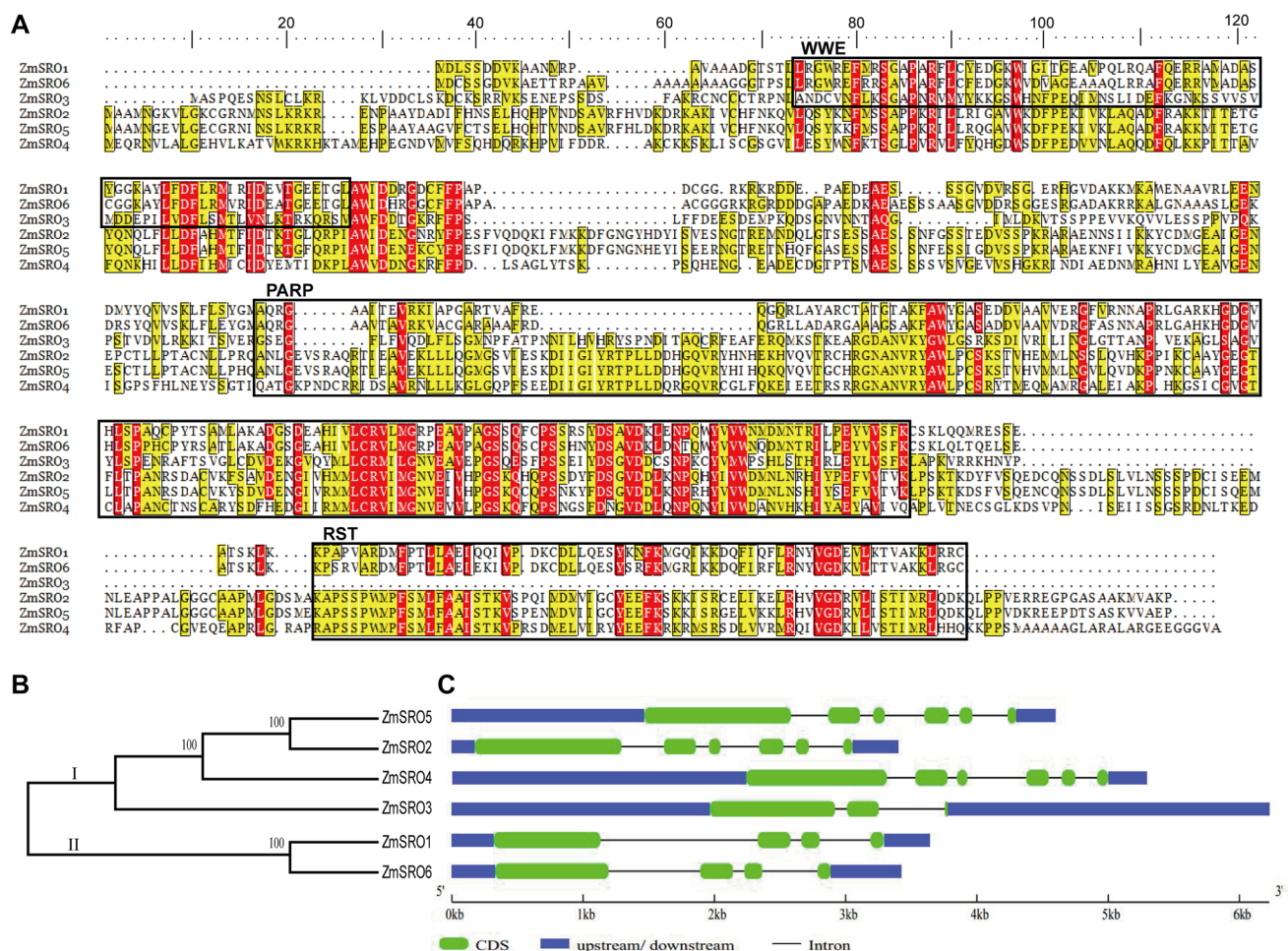


Fig. 1 Sequence alignment, phylogenetic analysis and gene structure analysis of ZmSROs. **a** Multiple sequence alignment of six maize SRO proteins. The PARP, WWE and RST domains were showed with boxes, respectively. **b** An un-rooted tree was constructed using

the full-length sequences of the six maize SRO proteins by the NJ method with 1000 bootstrap replicates. **c** The exons and introns were indicated by green boxes and gray lines, respectively. The untranslated regions were showed by blue rectangles

Table 1 The sequence characteristics of identified SROs in maize

Name	Sequenced ID	Size (aa)	MW (kDa)	PI	GRAVY	Chr.	Subcellular localization
ZmSRO1	GRMZM2G122543_T01	431	48.09	8.41	-0.465	1	Chloroplast
ZmSRO2	GRMZM5G866843_T03	596	66.78	8.23	-0.437	1	Nucleus
ZmSRO3	GRMZM2G011469_T01	404	45.32	8.14	-0.447	4	Nucleus
ZmSRO4	GRMZM2G072894_T01	580	64.79	7.01	-0.395	5	Nucleus
ZmSRO5	GRMZM2G145236_T02	596	66.65	8.02	-0.426	5	Nucleus
ZmSRO6	GRMZM2G177878_T02	448	48.04	9.17	-0.364	9	Cytoplasmic

including 4 exons and 3 introns. However, *ZmSRO3* only has two introns. By contrast, the gene structures of *ZmSROs* are consistent with the phylogenetic tree (Fig. 1). A total of 20 conserved motifs were identified in *ZmSRO* proteins, and all of them were then further annotated using the Pfam and SMART. As expected, the *ZmSROs* had a similar gene structure which also shared a similar motif composition. Motif

4 and 6 encode the RST domain, while motif 1 encodes the PARP domain (Fig. 2). Motif 2 is distributed in all the six sequences, encoding the Pleckstrin homology (PH)-like domain. Furthermore, motif 3 with the unknown functions was found in all protein sequences except for *ZmSRO1* and *ZmSRO6*. The detailed information of conserved amino acid sequences in the main motifs was shown in Table 2.

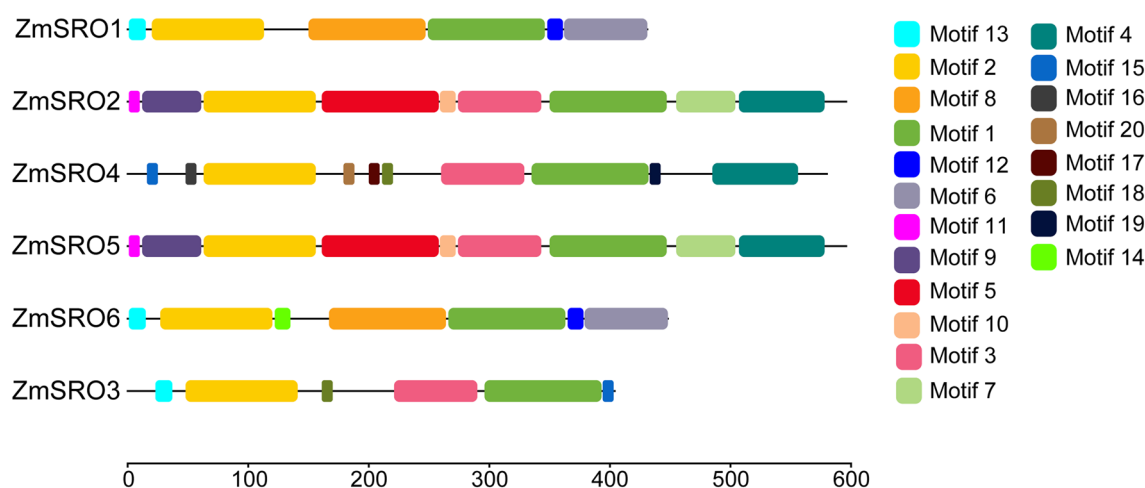


Fig. 2 Conserved motif analysis of ZmSRO proteins. The motifs in ZmSRO proteins were identified by the MEME. The different motifs were represented by different colors boxes. PARP domain: Motif1, RST domain: Motif 4 and Motif6, PH-like domain: Motif 2

Table 2 The major conserved motifs identified in ZmSROs by MEME analysis

Motif	Width	Conserved amino acid sequences
1	98	PPLKAAAYGEGVYLSANRPYTSVKLSVDENGIVHMLCRVIMGNVEAVPPGSKQFQPSSKYYDSGVDDLKN-PQHYVVWDMNLNTHIYPEYVSVKL
2	94	CHGNPQVLQSYKNFMSSAPPKRILLYQGGVWKDFPEKIVKLAQADFRAKKTITETGYQNKLFLLDFLHMTFIDTKT-GLQRPIAWIDDNGKRFFP
3	70	LLLQGMGSVIESKDIIIGIYRTPLDDHGQVRYHIHZKQVQVTRCHRGANVRYAWLPCSKSTVHQMMLNG
4	72	APSSPWMPFSMLFAAISTKVSPQBMDLVIGCYEEFKSKKISRGLVKKLRHVVGDRVLSTIMRLQDKQLPP
5	98	QDQKJFMKKDFGNGNHEYISEERNGTREMNDQFGAESSAESSNFESSIEDVSSPKRARAENFIKKYCDMGAEIGE-NEPCTLLPTACNLLPRQANL
6	70	LKKPARVARDMFPTLLAEIZKIVPDKCDLLQESYKRFKMGRICKDQFIRFLRNYVGDEVLTVAKKLRGC
7	50	FVSQEDCQNSSDLSLVLNSSSPDCISZEMNLEAPPALGGGCAAPMLGDSM
8	98	DAKKRKAWENAAYVLEEKDRYYQVVSCLFLEYGMAQRGAATEVRKIACGARAVAFREQRRRLADARCAAAGSAK-FAWYGASEDDVAAYVVERGFVRNN
9	50	CGRNINSLKRKRENPAAYDAGIFCNSELHQHPVNDASVRFHLDKDRKAKI
13	15	MDCSSGDVKAENMRP

Subcellular localization prediction and *cis*-elements analysis

The subcellular localization of ZmSRO proteins was predicted using the online tools WoLF PSORT and Plant-mPloc. Comprehensive analysis showed that ZmSRO1 is located in chloroplast, but ZmSRO2, 3, 4 and 5 were predicted in nucleus, and ZmSRO6 was predicted in cytoplasm (Table 1). The 2000 bp upstream promoter sequences of *ZmSROs* were submitted to the PlantCARE database to identify the putative *cis*-elements. Five stress-related *cis*-elements were detected, including the anaerobic induction element (ARE), MYB binding site involved in drought and stress (MBS), defense and stress-responsive elements (TC-rich repeats), heat shock element (HSE) and low temperature-responsive element (LTR) (Fig. 3). Additionally, the

elements that participated in response to hormones were also identified, including gibberellin-responsive element (GARE-motif and P-box), auxin-responsive element (TGA-element), salicylic acid-responsive element (TCA element), MeJA-responsive element (TGACG-motif or CGTCA-motif), ethylene-responsive element (ERE) and abscisic acid-responsive element (ABRE) (Fig. 3).

Phylogenetic analysis of SROs in different plants

To investigate the evolutionary relationship of ZmSROs, the full-length protein sequences were used to construct the phylogenetic tree. The six ZmSROs were divided into two groups (Fig. 1a). ZmSRO1 and ZmSRO6 fall into one group, while the others are shown in the same group. To further explore the evolutionary relationship among

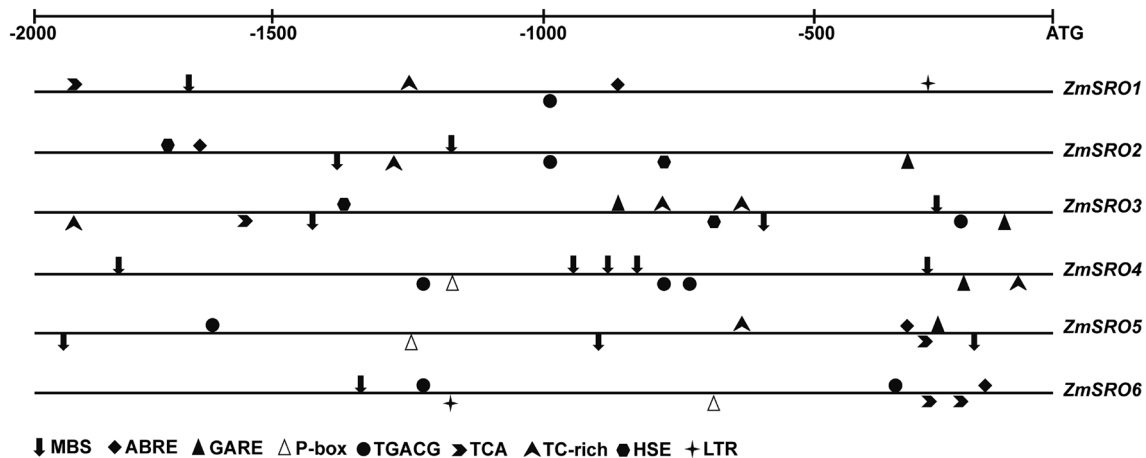


Fig. 3 The major stress-related and hormone-induced *cis*-elements were identified in the putative promoter sequences of the six *ZmSRO* genes. *MBS* MYB-binding site element, *ABRE* abscisic acid-responsive element, *GARE* gibberellin-responsive element, *P-box* gibberel-

lin-responsive element, *TGACG* TGACG-motif element, *TCA* TCA element, *TC-rich* TC-rich repeats element, *HSE* heat stress-responsive element, *LTR* low temperature-responsive element

different species, a total of 76 SRO protein sequences from 33 species including *Arabidopsis*, rice, wheat, *Brassica napus* and so on were identified from the public database for evolutionary tree (Fig. 4). These SRO proteins were divided into four groups (I, II, III and IV). Six *ZmSROs* were clustered in Group II, III and IV, containing two, one and three members, respectively. Interestingly, the members of group I were only found in dicots, and the members from group III were all from monocots. In the group II and group IV, these members come from dicots and monocots. By contrast, the maize SRO proteins had a more close relationship with rice SRO proteins.

Expression profiles of *ZmSROs* during developmental processes and different tissues of maize

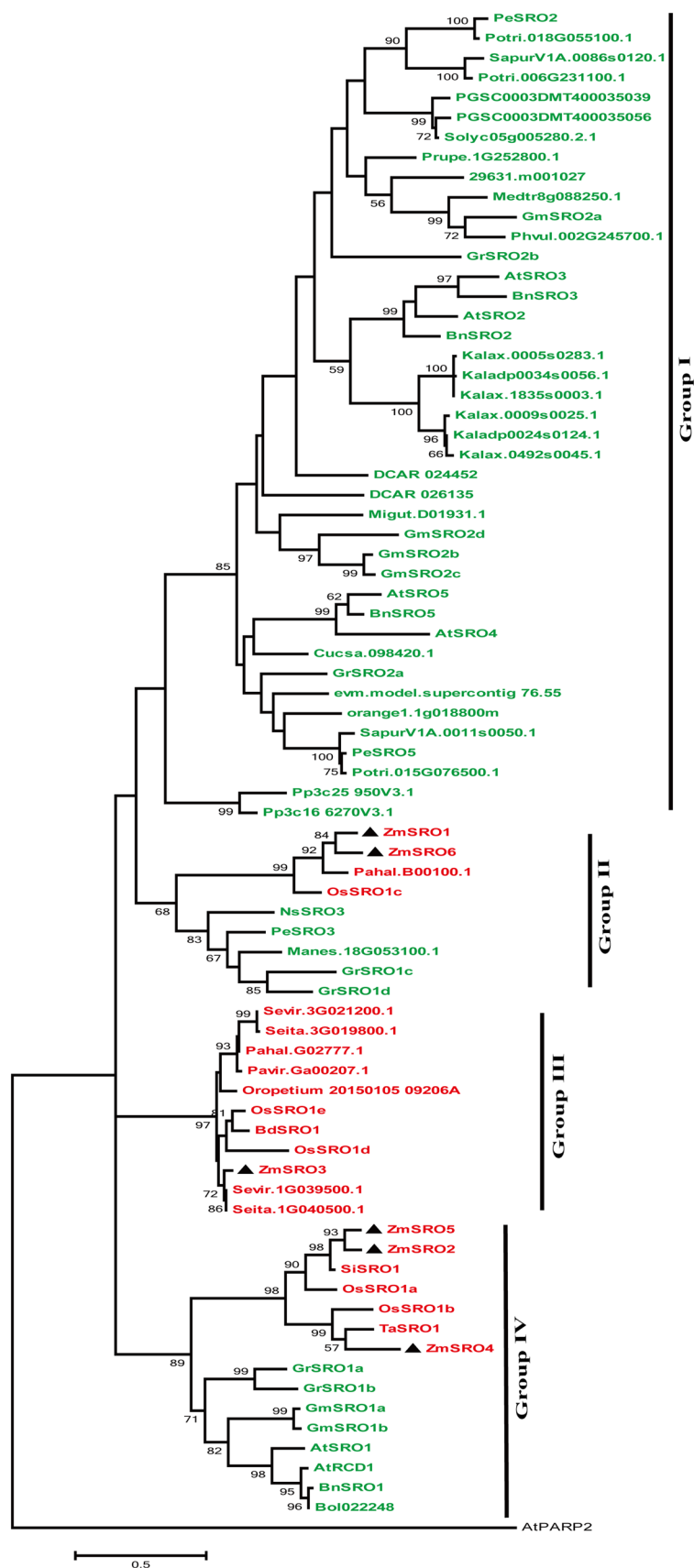
The publically available genome-wide transcript data from maize tissues at different developmental stages were used to investigate the expression patterns of the *ZmSROs* (Stelpflug et al. 2016). As showed in Fig. 5, the expressions of *ZmSROs* were detected in majority of tissues and developmental stages. *ZmSRO2* and *ZmSRO5* had a similar expression pattern, with high expression levels in primary root and internode, but low expression levels were exhibited in seed, embryo, and endosperm. Both *ZmSRO1* and *ZmSRO6* were expressed at relatively low levels in all the tissues and developmental stages except for old leaf. However, *ZmSRO1* exhibited a relatively high expression level in seed, while *ZmSRO3* and *ZmSRO4* had obvious differences at the different developmental stages of embryo and endosperm.

Expression profiles of *ZmSROs* under abiotic stress treatments

PEG6000 and NaCl treatments were used to simulate plant drought and salt stresses. Under the drought stress, *ZmSRO1* was repressed at 0.5 h, 2 h, 4 h and 6 h time points, but was induced and peaked at 8 h. *ZmSRO6* was repressed at all the five time points. The expression levels of *ZmSRO3* and *ZmSRO4* were up-regulated more than fourfold at 0.5 h, and then were down-regulated subsequently. *ZmSRO3* was restored to the control levels at 8 h, while *ZmSRO4* recovered to the control levels at 6 h and was again induced at 8 h. In addition to the no significant changes of *ZmSRO2* at 0.5 h and *ZmSRO5* at 2 h, both of them were repressed at all the five time points. For NaCl treatment, *ZmSRO1* and 6 were down-regulated at all the five time points. The expression peak of *ZmSRO3* was observed at 0.5 h, similar with PEG6000 treatment, and then was repressed, but up-regulated again at 8 h. *ZmSRO4* was induced at 0.5 h, 4 h and 8 h and peaked at 8 h. *ZmSRO5* was also induced and peaked at 8 h, but repressed at other time points. Although *ZmSRO4* and *ZmSRO5* showed the highest expression levels at 8 h, *ZmSRO4* was significantly higher than that of *ZmSRO5*. *ZmSRO2* was induced at 2 h and 4 h and peaked at 4 h, while down-regulated at 0.5 h and 8 h (Fig. 6).

ABA and GA3 were used to explore hormone response of *ZmSRO* genes. Under the ABA stress, *ZmSRO1* displayed no significant change in response to it at 1 h and 3 h, but was induced and peaked at 9 h and repressed at 6 h and 12 h. *ZmSRO6* was induced at 6 h and 9 h and peaked at 6 h, but repressed at other time points. The highest expression levels of *ZmSRO4* and *ZmSRO5* occurred at 3 h and were down-regulated after that point. *ZmSRO2* was significantly

Fig. 4 Phylogenetic relationship of SRO proteins in different plants. The un-rooted phylogenetic tree was constructed in MEGA6.0. AtPARP2 (accession number: AT4G02390) gene was used as an outgroup. The bootstrap values higher than 50% from 1000 replicates were shown beside the branches and the SRO proteins of monocotyledons and dicotyledons were shown in red and green, respectively. The protein sequences came from 33 species and the detailed information on the species and sequences were shown in Table S2



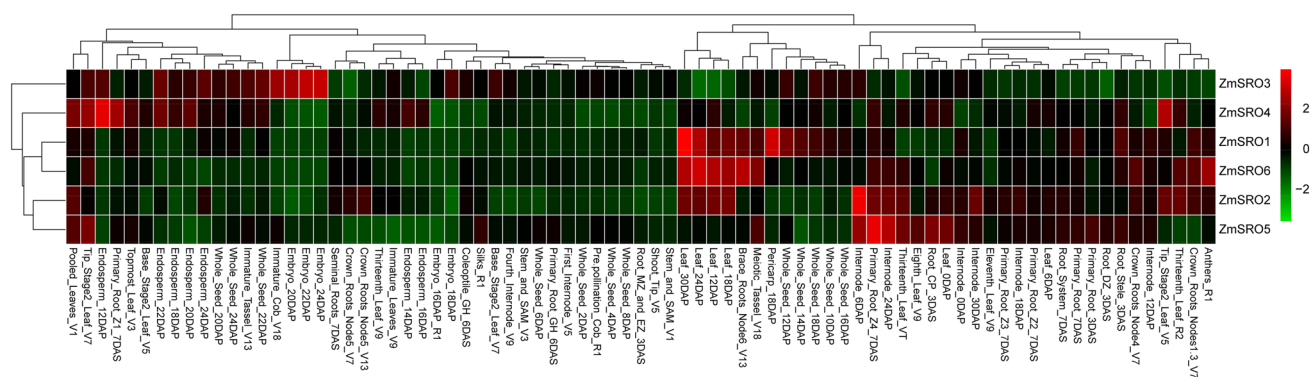


Fig. 5 The expression patterns of *ZmSROs* in different tissues of maize. The RNA-seq data were obtained from (<https://ftp.maizgedb.org/MaizeGDB/FTP/B73%20RNA-SEQ%20Gene%20Atlas/>) pro-

vided by Stelpflug et al. (2016). The expression data were gene-wise normalized and used to generate a heatmap that was performed using a online platform of OmicShare (<http://www.omicshare.com/tools>)

repressed at 1 h, 6 h and 12 h, but induced at 3 h and 9 h and peaked at 9 h. *ZmSRO3* were down-regulated at all the five time points. Under the GA3 stress, the highest expression levels of *ZmSRO1*, 3, 4 and 6 were observed at 9 h after treatment, while *ZmSRO2* and *ZmSRO5* occurred at 6 h. *ZmSRO6* induced by GA3 was higher than that induced by ABA (Fig. 6).

Discussion

In our present study, six *ZmSROs* were identified in maize genome. A comprehensive analysis including gene and protein structures, conserved motifs, subcellular localization, *cis*-regulatory elements, phylogeny and expression profiles were carried out to characterize the *ZmSRO* gene family.

The analyses of gene structure and conserved motifs indicated that *ZmSRO* gene family could be divided into two groups (Figs. 1, 2). Within the same group, *ZmSROs* shared similar conserved domains. In *Arabidopsis*, the SRO gene family was divided into two types based on whether they had an N-terminal WWE domain (Jaspers et al. 2010). We found that *ZmSRO1*, 3 and 6 contained the WWE domain, but *ZmSRO3* did not have the RST domain. Unsurprisingly, *ZmSRO1* and 6 were clustered together. *ZmSRO3* was grouped into other members. In rice, *OsSRO1a* to *OsSRO1e* had the WWE domain, but *OsSRO1c* and *AtSRO2* to *AtSRO5* were clustered in the same group (You et al. 2013). The result indicated that the RST domain was more conservative than the WWE domain. A previous study showed that different SRO groups closely related to different plant groups during evolution (Jaspers et al. 2010). In this study, the phylogenetic analysis of SROs from 33 species showed that the same sub-groups did not differentiate the members derived from both dicotyledons and monocotyledons (Fig. 4), which revealed that the genome evolution of SROs

followed the monocot and dicot divergence. The phylogenetic analysis also showed that *ZmSROs* were more close to rice homologs, suggesting the functional similarity of SROs in two crops. Generally speaking, the patterns of gene expression reflect their functions. The public databases provide us with a lot of useful information, basing on the available public data to explore the expression patterns of genes, especially gene families, has been applied in many studies (Venter and Botha 2004; Chen et al. 2014; Li et al. 2017a). To explore the functions of *ZmSROs*, the publically available RNA-seq data from 79 different tissues and developmental stages of the maize line B73 were used as a main source for analyzing transcriptional expression patterns of SROs (Stelpflug et al. 2016). The expression patterns of them not only supported the phylogenetic reconstruction, but also suggesting their functional diversity. In addition, *ZmSRO2* and *ZmSRO5*, *ZmSRO1* and *ZmSRO6*, as well as *ZmSRO3* and *ZmSRO4* had the similar expression patterns, respectively, which implied the possible functional redundancy in maize development.

The *cis*-regulatory elements in the promoter region are important molecular switches for gene expression (Hernandez-Garcia and Finer 2014). In the previous studies, some specific promoters in regulation of gene transcription under abiotic stress had been identified (Lee et al. 2010; Yao et al. 2017). In this study, the promoter sequences of *ZmSROs* were predicted and the *cis*-elements were searched against the PlantCARE database, respectively (Lescot et al. 2002; Li et al. 2016). Some increasing evidence suggested that SRO gene played a crucial role in response to the different stresses and hormone induction (Babajani et al. 2009; You et al. 2013; Liu et al. 2014; Anjum et al. 2015). Several *cis*-acting regulatory elements, such as ARE, MBS, TC-rich repeats, HSE and LTR were identified from the promoter regions of different *ZmSROs* (Fig. 3) (Dey et al. 2015). Moreover, the hormone-induced elements by gibberellin, jasmonic acid,

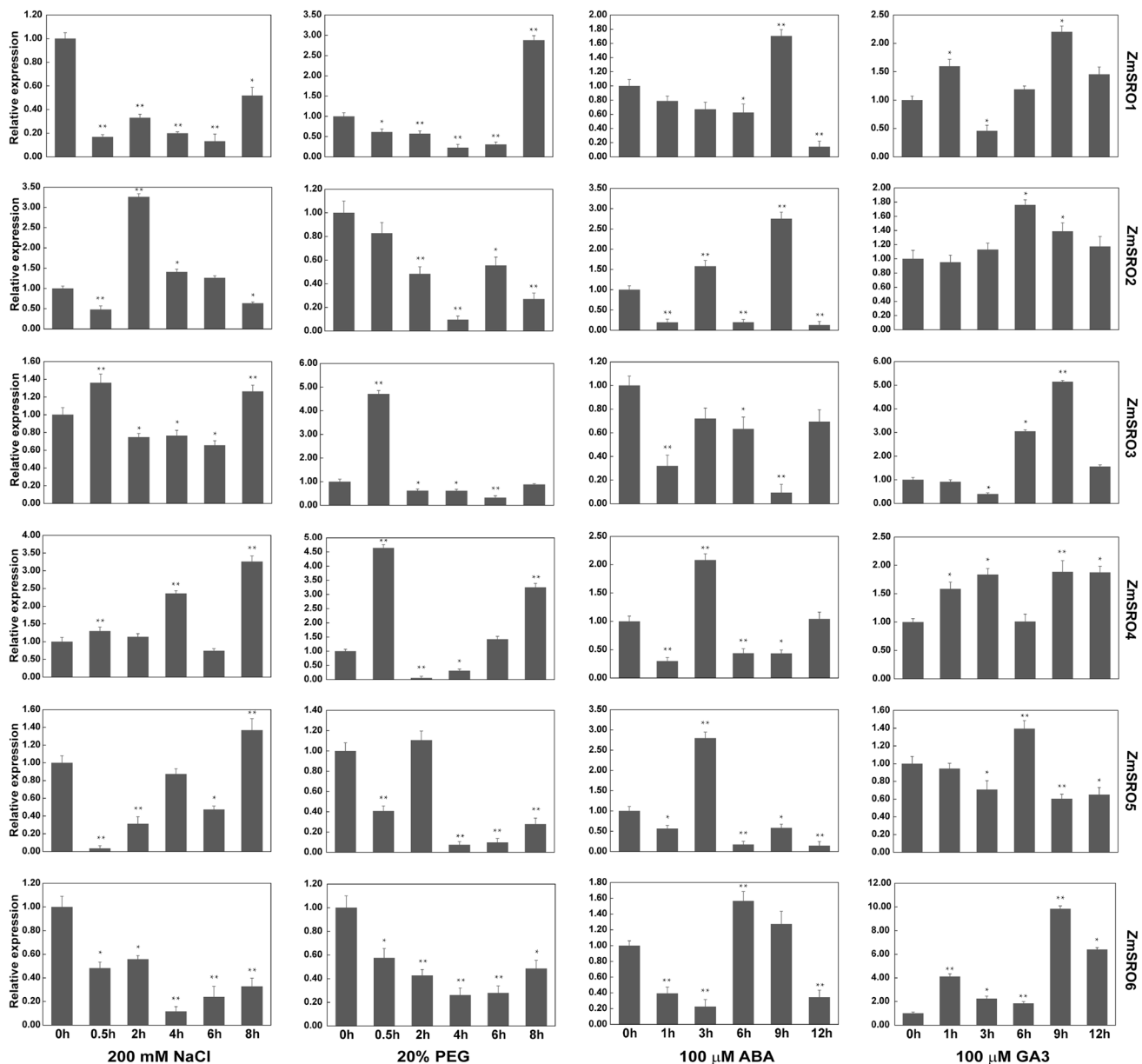


Fig. 6 Expression patterns of *ZmSROs* in response to PEG, NaCl, exogenous ABA and GA treatments. The Y-axis represents the relative expression levels and the X-axis shows the time points of

stresses treatment. Bars represent mean ± SE ($n=3$). Significant differences from the 0 h were determined by Student's *t* test (* $p < 0.05$, ** $p < 0.01$)

auxin, ethylene and abscisic acid were also identified. It implied that the functions of *ZmSROs* were consistent with the *SRO* gene family in other species. These results suggested that they were not only regulated by the abiotic stress, but also were induced by the various hormones. However, the actual function of each *ZmSRO* member requires further research.

Although a growing numbers of *SROs* had been identified from a variety of plants, their functions were still limited to a few of model plants, such as *RCD1*, *SRO1* and *SRO5* of *Arabidopsis* and *OsSRO1c* of rice. Recent studies had

confirmed the special functions of *SROs* in plant growth and development as well as in response to abiotic stress and hormones (Katiyar-Agarwal et al. 2006; Teotia et al. 2010; Vainonen et al. 2012; You et al. 2013; Liu et al. 2014). However, no data of *ZmSROs* functions in response to drought, salt and hormones stresses are reported in maize. In this study, the PEG6000, NaCl, exogenous ABA and GA3 were used to deal with maize seedlings, respectively. Results from qPCR analysis confirmed that *ZmSROs* were indeed regulated by drought, salt, ABA and GA (Fig. 6). In general, genes with similar structure or relationship have similar

functions (Guo et al. 2017). However, *ZmSROs*, the closely related members did not always have the similar expression patterns, although they shared many similar characteristics. For example, *ZmSRO1* and *ZmSRO6* were in the same group, and they shared the similar motifs in amino acids sequences and *cis*-regulatory elements in promoter regions. However, the expression levels of *ZmSRO1* were induced by drought and repressed by salt, while *ZmSRO6* was repressed by both. Moreover, they had the highest homology with *OsSRO1c*, but *OsSRO1c* was strongly regulated by drought and salt stresses (You et al. 2013). Gene expression regulation were affected by the gene promoters (Yao et al. 2017; Venter and Botha 2004). In this study, *ZmSRO3* and 4 were significantly induced by drought, which might be related to the fact that they contained more MBS element in the promoter region. However, the expressions of *ZmSRO2* and 6 were inhibited in response to drought, although both of them had the MBS element in their promoter region. In addition, all of the six genes were induced in response to ABA and GA stress except for *ZmSRO3*, which might be related to the absence of ABRE in the promoter region of *ZmSRO3*. Therefore, gene functions were influenced by the *cis*-regulatory elements, but did not depend on these *cis*-regulatory elements. Interestingly, *ZmSRO3* had the longest genomic sequence among the six *SRO* genes, but had the least number of amino acids. *ZmSRO3* was the only one that did not have the RST domain and was also the only one that down-regulated in response to ABA stress at all the five time points in this study. Therefore, *ZmSRO3* needed special attention in subsequent studies. It also should be noted that the expression of genes is not a simple linear relationship with the time change under different treatments, and this phenomenon has been displayed in other studies (Cheng et al. 2017; Wang et al. 2015; Li et al. 2017a, b). These indicate the complexity of gene regulation. Although our results preliminarily suggest that the *ZmSRO* genes do respond to abiotic stress, there is still a lot of work to be done in the future to resolve the response mechanism.

In conclusion, a total of six maize *SRO* genes were identified from the maize genome, and the sequences of them were characterized. First, *ZmSROs* expressed in different tissues during maize growth and development played a vital role in the developmental processes of maize as in other plants. Meanwhile, *ZmSROs* could be significantly responsive to abiotic stress and hormones, and the different members exhibited different patterns. Our study provided a basis for studying the functions of *SRO* gene family of maize, especially for the family members with potential functions in response to drought and salt stress.

Author contribution statement HJ and YX performed the experiments, data analyses and manuscript preparation, together. SZ was responsible for the whole study and the manuscript preparation.

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Compliance with ethical standards

Conflict of interest All authors declared that they had no conflict of interest.

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